References and Mathematical Equations for the $\mathcal{N} = 2$ Lagrangian Gauge-Theory Utilities

Lie algebra data, anomaly cancellation, beta functions, flavor symmetry, and conformal-manifold dimension

This document records the scholarly and software references used in the implementation, states the mathematical conventions, and associates each implemented equation with a source whenever possible. Equations marked “derived” are direct consequences of the cited definitions rather than formulas copied from an external source.

SuperconformalIndex project documentation

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Contents

1 Purpose and implementation map

The current implementation has three mathematical layers:

1. SageMath-backed finite simple Lie-algebra and representation data;
2. anomaly and one-loop beta-function checks for simple and semisimple gauge groups;
3. connected flavor symmetry and local conformal-manifold dimension for Lagrangian SCFT candidates.

The principal implementation locations are:

Module	Responsibility
`lie_algebra.py`	Cartan types, roots and weights, irreducible characters, dimensions, duals, Casimirs, indices, and representation reality.
`check_n2_anomalies.py`	Hypermultiplet validation, perturbative anomaly argument, Witten parity, spectator dimensions, and beta functions.
`n2_theory_properties.py`	Connected continuous flavor symmetry and the local dimension of the conformal manifold.

The original SU(N)-only source and tests came from the internal handoff document [1]. Its Young-diagram formulas were later generalized to all finite simple Cartan types by replacing the hand-coded group theory with SageMath.

2 Reference roles

Reference	Role in the implementation or its validation
Internal handoff [1]	Starting SU(N) implementation, exact arithmetic, input schema, tests, beta functions, spectator dimensions, and SU(2) parity.
SageMath root systems [2]	Computational source for finite Cartan types, roots, coroots, weights, highest roots, and Bourbaki numbering.
SageMath Weyl characters [3]	Computational source for irreducible characters, degrees, dual characters, highest weights, and Frobenius-Schur indicators.
Bhardwaj–Tachikawa [4]	Main physics source for $\mathcal{N}=2$ hypermultiplets, semisimple product representations, one-loop beta functions, spectator dimensions, and symplectic global-anomaly integrality.
Witten [5]	Original source for the conventional four-dimensional SU(2) global anomaly.
Wang–Wen–Witten [6]	General SU(2)-representation parity, the symplectic analogue, and the distinction between the conventional spin-manifold anomaly and the newer non-spin anomaly.
García-Etxebarria–Montero [7]	Spin-bordism validation of which supported simply connected simple groups can carry the relevant four-dimensional global gauge anomaly.

Reference	Role in the implementation or its validation
Córdova–Dumitrescu–Intriligator [8]	Protection of marginal deformations preserving four-dimensional $\mathcal{N}=2$ supersymmetry and hence the exactly marginal status of conformal gauge couplings.
LieART 2.0 [9]	Cross-check of representation dimensions, indices, and exceptional representations. It is not an implementation backend because its exceptional-node convention differs from SageMath.
Bilal [10]	Supporting reference for the perturbative non-abelian cubic anomaly and its behavior under complex conjugation. Added for equation-level attribution; it is not a software dependency.
Tachikawa [11]	General supporting background for four-dimensional $\mathcal{N}=2$ gauge theories, flavor symmetries, and complexified gauge couplings.

3 Lie algebra and representation theory

3.1 Dynkin labels and highest weights

For simple roots α_i, simple coroots $\alpha_i^\vee$, and fundamental weights ω_i, the Dynkin labels of a highest weight λ are

$$a_i = \langle \lambda, \alpha_i^\vee \rangle = \frac{2(\lambda, \alpha_i)}{(\alpha_i, \alpha_i)}, \qquad \lambda = \sum_{i=1}^{r} a_i \omega_i. \tag{1}$$

Equation source: SageMath root-system conventions [2]; LieART Eq. (13) [9].

This is the mathematical operation performed by `_irreducible_character`: validated input labels are converted to a dominant highest weight, then passed to SageMath's Weyl character ring.

If θ is the highest root, the adjoint Dynkin labels are

$$a_i^{\mathrm{adj}} = \langle \theta, \alpha_i^\vee \rangle. \tag{2}$$

Derivation/source: The adjoint highest weight is the highest root; implemented using SageMath's highest root and simple coroots [2].

3.2 Dimensions

For rank r and root system Δ, the algebra decomposes into its Cartan subalgebra and one-dimensional root spaces, so

$$\dim \mathfrak{g} = r + |\Delta|. \tag{3}$$

Derivation/source: Standard root-space decomposition; the root set is supplied by SageMath [2].

For a highest-weight irrep V_λ, its dimension is given mathematically by the Weyl dimension formula

$$\dim V_\lambda = \prod_{\alpha\in\Delta_+} \frac{(\lambda+\rho,\alpha)}{(\rho,\alpha)}, \qquad \rho = \frac{1}{2}\sum_{\alpha>0}\alpha. \tag{4}$$

Equation source: LieART Eq. (14) [9]; SageMath returns the same quantity through the character degree [3].

3.3 Quadratic Casimir and Dynkin index

Let long roots have length squared $(\theta,\theta) = 2$. The code uses

$$C_2(R) = \frac{(\lambda,\lambda+2\rho)}{(\theta,\theta)}, \qquad C_2(\text{adj}) = h^\vee. \tag{5}$$

Equation source: LieART scalar-product convention and Eq. (15) [9]; the division by the long-root length fixes the code's $C_2(\text{adj}) = h^\vee$ convention.

The Dynkin index is then

$$\text{Tr}_R(T^aT^b) = T(R)\delta^{ab}, \qquad T(R) = \frac{\dim R}{\dim \mathfrak{g}}C_2(R). \tag{6}$$

Equation source: Bhardwaj–Tachikawa Eq. (2.5) and surrounding definitions [4]; LieART Eq. (15) [9].

This normalization gives

$$T(\mathbf{N}\text{ of }SU(N)) = \frac{1}{2}, \qquad T(\text{adj}) = h^\vee. \tag{7}$$

Derivation/source: Substitution into Eqs. (5)–(6); it matches the beta-function normalization in [4].

3.4 Duals and representation reality

SageMath classifies an irreducible character with the Frobenius-Schur indicator

$$\nu(R) = \int_G \chi_R(g^2)\,dg = \begin{cases} +1, & R \text{ real},\\ 0, & R \text{ complex},\\ -1, & R \text{ pseudoreal (quaternionic)}. \end{cases} \tag{8}$$

Equation source: SageMath Weyl-character documentation for `frobenius_schur_indicator` [3].

For an external tensor product of irreducible representations of independent group factors,

$$\nu(\boxtimes_{a=1}^s R_a) = \prod_{a=1}^s \nu(R_a). \tag{9}$$

Derivation/source: The invariant bilinear form on the tensor product is the tensor product of the factor forms. This is also the reality rule used in Bhardwaj–Tachikawa Eq. (2.3) and its discussion [4].

Thus a product is complex if any factor is complex. Otherwise it is pseudoreal exactly when an odd number of factors are pseudoreal.

3.5 SageMath and LieART conventions

LieART's integer index is

$$\ell_{\text{LieART}}(R) = \frac{\dim R}{\dim \mathfrak{g}}(\lambda, \lambda + 2\rho) = 2T_{\text{code}}(R). \tag{10}$$

Equation source: LieART Eq. (15) [9], compared with Eqs. (5)–(6).

The exceptional generating representations also use different node labels:

Representation	SageMath/Bourbaki	LieART
E_7 representation **56**	(0000001)	(0000010)
E_8 representation **248**	(00000001)	(00000010)

Equation source: SageMath type-E diagrams [2]; LieART Table 5.2 and Tables A.58–A.59 [9].

Consequently, LieART is used only for cross-checking after translating conventions.

4 Hypermultiplets and one-loop beta functions

4.1 Full and half hypermultiplets

For a semisimple gauge group and an irreducible matter block,

$$G = \prod_{a=1}^{s} G_a, \qquad R_h = \boxtimes_{a=1}^{s} R_{h,a}. \tag{11}$$

Equation source: Bhardwaj–Tachikawa Eqs. (2.1)–(2.3) [4].

A full hypermultiplet in R has the gauge representation

$$R \oplus \overline{R}, \tag{12}$$

while a proper half hypermultiplet is allowed only when R is pseudoreal.

Equation source: Bhardwaj–Tachikawa Eq. (2.4) and preceding discussion [4].

For factor G_a, the dimensions of all other representations act as a multiplicity:

$$d_{h,\widehat{a}} = \prod_{b \neq a} \dim R_{h,b}. \tag{13}$$

Equation source: Bhardwaj–Tachikawa Eqs. (2.5), (2.7), and (2.8) [4].

4.2 Beta function

Writing $c_h = 2$ for a full hyper and $c_h = 1$ for a half hyper, the implemented coefficient for each simple factor is

$$b_{0,a} = 2h_a^\vee - \sum_h n_h c_h T_a(R_{h,a}) \prod_{b\neq a} \dim R_{h,b}. \tag{14}$$

Equation source: Bhardwaj–Tachikawa Eqs. (2.5)–(2.8) [4].

For a single factor this becomes

$$b_0 = 2h^\vee - \sum_{h\,\text{full}} 2n_h T(R_h) - \sum_{h\,\text{half}} n_h T(R_h). \tag{15}$$

Equation source: Specialization of Eq. (14); Bhardwaj–Tachikawa Eqs. (2.6)–(2.8) [4].

The implementation calls the theory a Lagrangian SCFT candidate when every implemented anomaly check passes and

$$b_{0,a} = 0 \qquad \text{for every simple factor } a. \tag{16}$$

Derivation/source: The code's candidate criterion; vanishing $b_{0,a}$ is the conformal case of the one-loop condition in [4].

5 Gauge anomalies

5.1 Perturbative cubic anomaly

For a left-handed Weyl fermion in R, the group-theoretic cubic tensor is

$$d_R^{abc} = \mathrm{Tr}_R\Big(T^a\{T^b, T^c\}\Big)\,. \tag{17}$$

Equation source: Bilal's non-abelian gauge-anomaly discussion [10].

In the conjugate representation,

$$T^a_{\overline{R}} = -(T^a_R)^T, \qquad d^{abc}_{\overline{R}} = -d^{abc}_R. \tag{18}$$

Equation source: Standard conjugate-representation identity; see Bilal [10].

Therefore the two Weyl fermions in a full hyper satisfy

$$d^{abc}_{R\oplus\overline{R}} = d^{abc}_R + d^{abc}_{\overline{R}} = 0. \tag{19}$$

Derivation/source: Immediate consequence of Eqs. (12) and (18); the full-hyper content is from [4].

If R is real or pseudoreal, then $R \simeq \overline{R}$. Hence

$$d_R^{abc} = d_{\overline{R}}^{abc} = -d_R^{abc} \quad \Longrightarrow \quad d_R^{abc} = 0. \tag{20}$$

Derivation/source: Self-conjugacy plus Eq. (18); see also the anomaly-free real/pseudoreal discussion in [10].

This proves the perturbative anomaly cancellation used by the program. The code does not calculate an independent cubic index: full hypers are vectorlike, allowed half hypers are pseudoreal, and adjoint gauginos are real.

Mixed local anomalies involving different simple factors contain

$$\mathrm{Tr}_{R_a} T_a^A = 0 \tag{21}$$

and therefore vanish for semisimple gauge groups.

Derivation/source: Generators of a finite-dimensional representation of a simple Lie algebra are traceless; standard group-theoretic reduction of the triangle anomaly [10].

5.2 Conventional Witten anomaly

For each SU(2) or symplectic gauge factor, the code evaluates

$$\mathcal{W}_a = \sum_{h\,\text{half}} n_h \left[2T_a(R_{h,a})\right] \prod_{b \neq a} \dim R_{h,b} \pmod 2. \tag{22}$$

Equation source: Witten [5]; Wang–Wen–Witten Sec. 2.3 and Eq. (2.9) [6]; spectator multiplicities from [4].

Anomaly freedom requires

$$\mathcal{W}_a = 0. \tag{23}$$

Equation source: The conventional mod-two anomaly-cancellation condition [5, 6].

For the SU(2) isospin-j representation,

$$2T(j) = \frac{2}{3} j(j+1)(2j+1), \tag{24}$$

which is odd precisely when

$$j = 2r + \frac{1}{2}, \qquad r \in \mathbb{Z}_{\geq 0}. \tag{25}$$

Equation source: Wang–Wen–Witten Eq. (2.9) and the following paragraph [6].

The code therefore marks A_1, every C_r, and the isomorphic $B_2 = C_2$ case as potentially carrying the conventional parity obstruction. In spin bordism, the relevant groups include

$$\Omega_5^{\text{Spin}}(BSU(2)) \cong \mathbb{Z}_2, \qquad \Omega_5^{\text{Spin}}(BSp(n)) \cong \mathbb{Z}_2, \tag{26}$$

The other supported simply connected simple cases have no corresponding degree-five obstruction in the stated scope.

Equation source: García-Etxebarria–Montero Sec. 3.2 and Table 4 [7].

The newer SU(2) anomaly for representations $j = 4r + \frac{3}{2}$ requires a non-spin formulation. Wang–Wen–Witten explicitly state that on an ordinary spin manifold there is no additional anomaly beyond the conventional one [6]; the implementation consequently excludes the newer anomaly from its scope.

6 Connected flavor symmetry

At the massless point, the connected continuous flavor group is the connected centralizer of the gauge action inside the group preserving the hypermultiplet's quaternionic symplectic form:

$$F_{\text{conn}} = C_{Sp(\mathcal{H})}(G)^{\circ}. \tag{27}$$

Derivation/source: Standard centralizer definition for hypermultiplet flavor symmetry; see Tachikawa's $\mathcal{N}=2$ gauge-theory lectures for background [11].

For each inequivalent irreducible total gauge representation, the multiplicity space gives

reality of the gauge representation	connected flavor factor
complex, n full hypers	$U(n)$
real, n full hypers	$Sp(n)$
pseudoreal, m half-hyper units	$SO(m)$

(28)

Derivation/source: Apply Eq. (27): a symmetric gauge-invariant form pairs with a symplectic multiplicity form, while an antisymmetric gauge form pairs with an orthogonal multiplicity form.

For a pseudoreal block, a full hyper consists of two half-hyper units, so

$$m = n_{\text{half}} + 2n_{\text{full}}. \tag{29}$$

Derivation/source: Full-hyper decomposition in Eq. (12); implemented directly in `n2_theory_properties.py`.

The dimensions and ranks reported by the program are

$$\dim U(n) = n^2, \qquad \operatorname{rank} U(n) = n, \tag{30}$$

$$\dim Sp(n) = n(2n+1), \qquad \operatorname{rank} Sp(n) = n, \tag{31}$$

$$\dim SO(m) = \frac{m(m-1)}{2}, \qquad \operatorname{rank} SO(m) = \left\lfloor \frac{m}{2} \right\rfloor. \tag{32}$$

Derivation/source: Standard classical Lie-group dimensions; the program uses these closed forms directly.

Complex-conjugate gauge representations are merged into one flavor block. The reported object is only the connected continuous group: finite center quotients and disconnected flavor transformations are not determined.

7 Conformal-manifold dimension

For each conformal simple gauge factor, the complexified coupling is

$$\tau_a = \frac{\theta_a}{2\pi} + \frac{4\pi i}{g_a^2}. \tag{33}$$

Equation source: Standard $\mathcal{N} = 2$ gauge-coupling convention; see Tachikawa [11].

Córdova–Dumitrescu–Intriligator show that, in four dimensions, every marginal deformation preserving $\mathcal{N} = 2$ supersymmetry is exactly marginal because its multiplet is absolutely protected [8]. Therefore, for a Lagrangian SCFT candidate with s simple conformal gauge factors,

$$\dim_{\mathbb{C}} \mathcal{M}_{\text{conf}} = s. \tag{34}$$

Equation source: Córdova–Dumitrescu–Intriligator Sec. 3.2.2 and Table 22 [8], together with one coupling (33) per simple gauge factor.

A discrete S-duality quotient can identify points but does not change this local complex dimension. For a future non-Lagrangian implementation, exactly marginal operators must be obtained from protected SCFT data rather than merely counting Lagrangian gauge factors.

8 Symplectic notation and global scope

The project uses the convention

$$Sp(n) = \text{compact symplectic group of rank } n, \qquad \dim(\text{defining representation}) = 2n. \tag{35}$$

Equation source: This is the project convention. Bhardwaj–Tachikawa write $USp(2n)$ [4], while Wang–Wen–Witten and LieART use dimension-based $Sp(2n)$ notation [6, 9].

The checker assumes the simply connected compact group associated with each finite simple Cartan type:

$$A_r \mapsto SU(r+1), \quad B_r \mapsto Spin(2r+1), \quad C_r \mapsto Sp(r), \quad D_r \mapsto Spin(2r). \tag{36}$$

Equation source: Standard finite Cartan classification and SageMath/Bourbaki conventions [2].

Current exclusions are:

- non-simply-connected quotient groups and line-operator choices;
- abelian gauge factors;
- higher-form and quotient-dependent global anomalies;
- the newer non-spin SU(2) anomaly;
- finite/disconnected flavor-group information;
- central charges, Coulomb-branch spectra, and superconformal indices.

Pure flavor and R-symmetry ’t Hooft anomalies are physical observables of global symmetries; unlike gauge anomalies, they are not required to cancel.

A Relation to the original SU(N) handoff formulas

The original handoff represented an SU(N) highest weight by Young-diagram row lengths

$$\ell_i = \sum_{k=i}^{N-1} a_k \quad (i = 1, \ldots, N-1), \qquad \ell_N = 0. \tag{37}$$

Equation source: Internal handoff mathematical specification [1].

It evaluated

$$\dim R = \prod_{1 \le i < j \le N} \frac{\ell_i - \ell_j + j - i}{j - i}, \tag{38}$$

Equation source: Internal handoff [1]; this is the type-A specialization of the Weyl formula (4).

With $B = \sum_i \ell_i$, its quadratic Casimir was

$$C_2(R) = \frac{1}{2}\left[NB + \sum_{i=1}^{N} \ell_i(\ell_i + 1 - 2i)\right] - \frac{B^2}{2N}, \tag{39}$$

Equation source: Internal handoff [1]; equivalent to Eq. (5) for type A_{N-1}.

and then used

$$T(R) = \frac{\dim R\, C_2(R)}{N^2 - 1}. \tag{40}$$

Equation source: Internal handoff [1]; type-A case of Eq. (6).

These formulas remain useful regression checks, but the current implementation obtains the same invariants uniformly from SageMath for classical and exceptional Lie algebras.

References

[1] Internal project handoff, *Handoff: 4d N=2 SU(N) anomaly and beta-function checker*, local project document.

[2] The SageMath Developers, *Root systems - SageMath Reference Manual*, `https://doc.sagemath.org/html/en/reference/combinat/sage/combinat/root_system/root_system.html`. The project was verified with SageMath 10.7; the linked reference manual may display a newer release.

[3] The SageMath Developers, *Weyl character rings - SageMath Reference Manual*, `https://doc.sagemath.org/html/en/reference/combinat/sage/combinat/root_system/weyl_characters.html`.

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